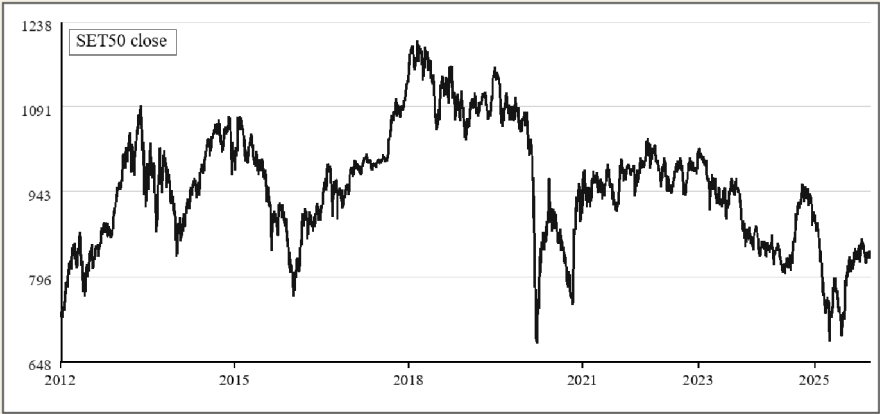


Input data

MARKET SERIES

SET50 daily index

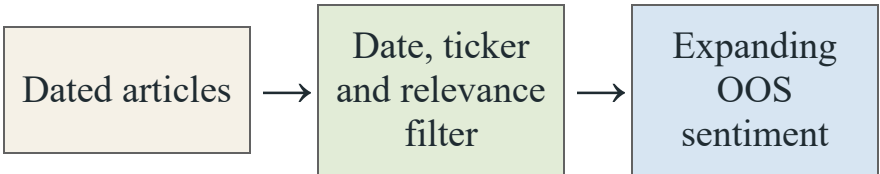
2012-2025 | 116 TA features | t+1 target



All numerical features are reconstructed using only information available by the close of day t.

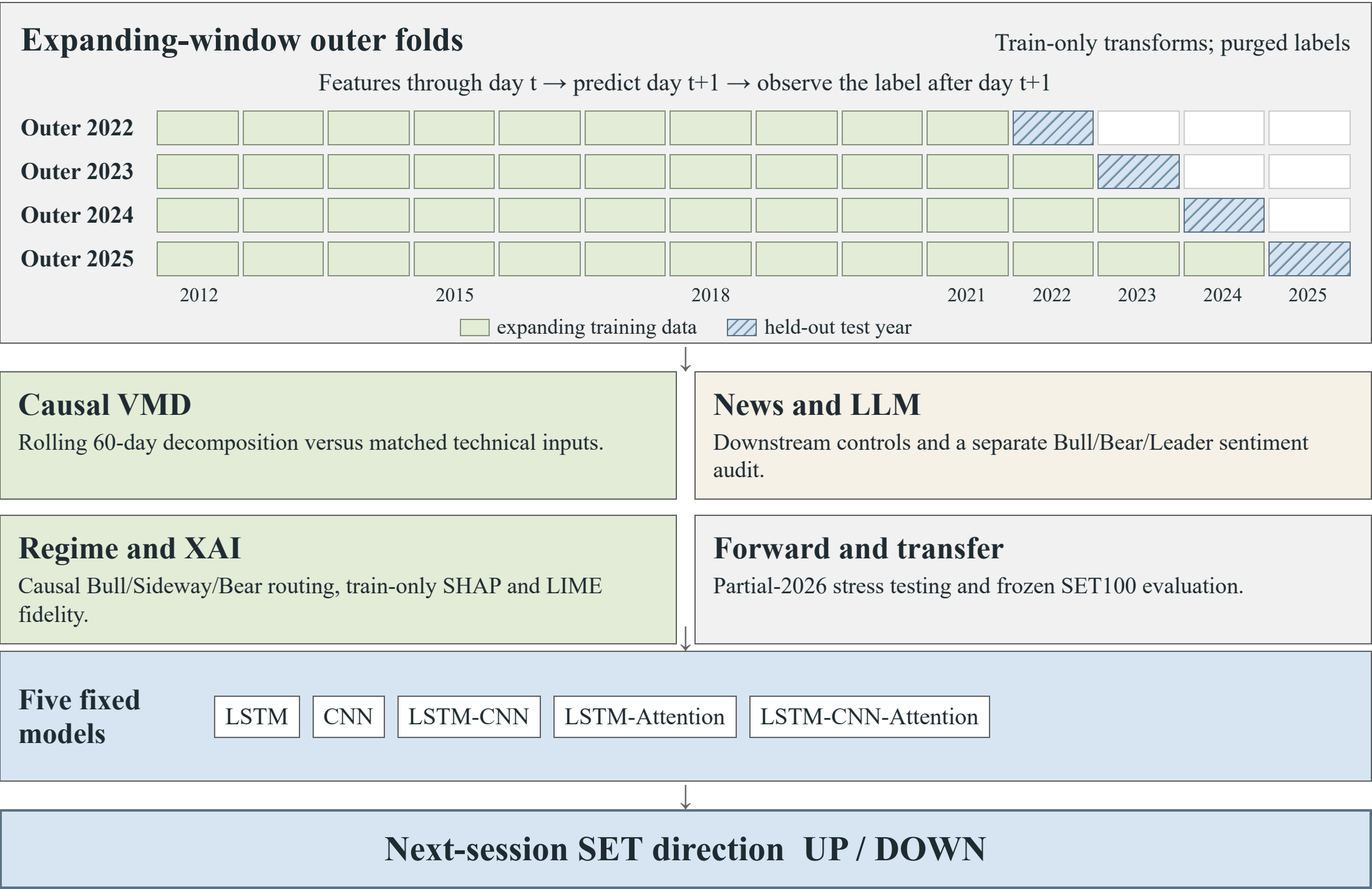
FINANCIAL-NEWS MODALITY

Dated news to out-of-sample sentiment



Sentiment is generated without using labels from the subsequent forecasting period.

Point-in-time reliability evaluation



Main findings

Best mean balanced accuracy
LSTM-CNN-Attention; four held-out years and five seeds.
53.64%

CNN with Regime-SHAP
Largest selective forecasting gain.
+1.46 pp

Leader sentiment
Versus equal-call and near-cost controls; intrinsic endpoint.
+5.93 / +6.00 pp

LIME fidelity stress test
Share of repeated explanations classified as low fidelity.
71.83%

Robustness results
VMD effect: -0.60 to +0.35 percentage points.
Predicted news: no consistent adjusted forecasting gain.
Partial-2026 stress test: one-sided collapse in several objectives.
SET100 transfer: lower mean balanced accuracy for all five models.
Observed improvements were architecture- and endpoint-dependent rather than universal.